

SDA Fall 2025 – Final paper solutions

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans	C	D	C	B	B	B	B	C	B	A	B	C	C	B	B	B	C	B	D	A

Q2: Homogenization: 1. Combining Classes

1. StudentInfo + StudentProfile + LearnerRecord → StudentRecord
2. CourseCatalog + SubjectList → CourseCatalog
3. TeacherAccount + FacultyProfile → FacultyProfile

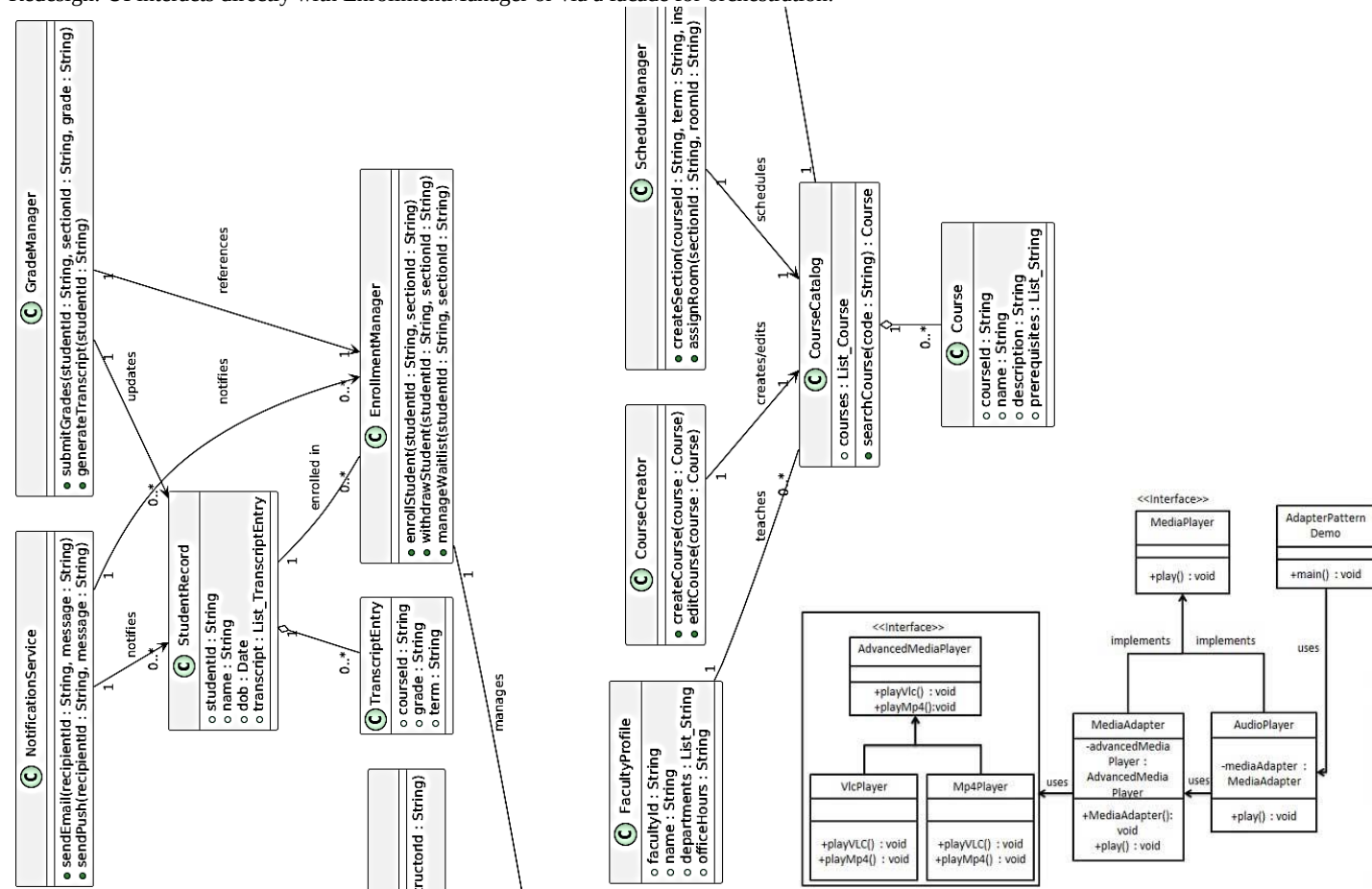
2. Splitting Classes (from overloaded CourseManagement)

1. CourseCreator – create/edit courses
2. SchedulerManager – manage sections and timetables
3. EnrollmentManager – enroll/withdraw students, manage waitlists

4. GradeManager – grade submissions and transcripts
5. NotificationService – send messages to students/faculty

3. Eliminating Classes: EnrollmentController removed: Reason: Pass-through with no sequencing or validation

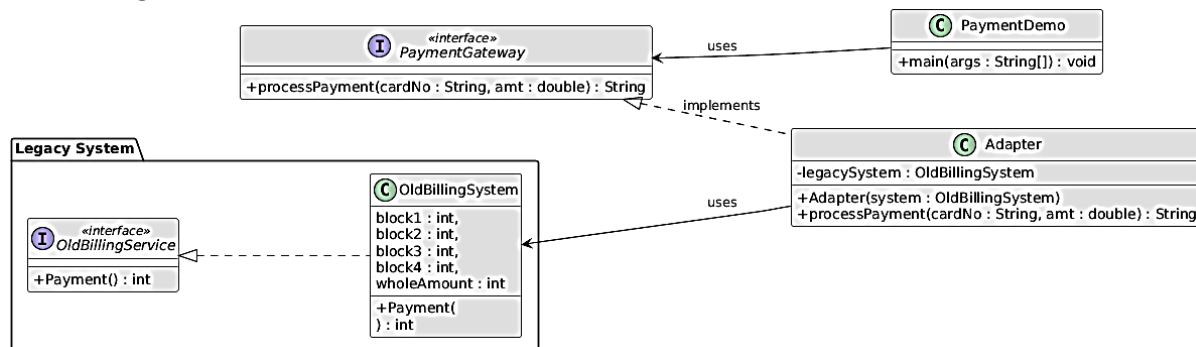
Redesign: UI interacts directly with EnrollmentManager or via a facade for orchestration.



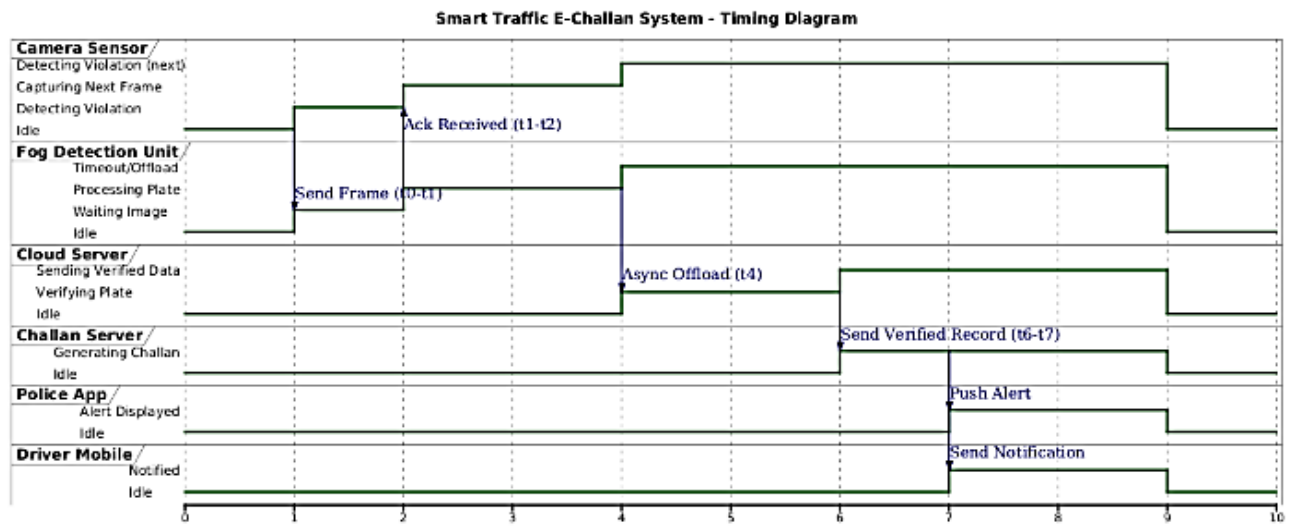
Q3

1. **Correct Design Pattern:** is the Adapter Design Pattern, because the new interface and legacy system have incompatible method signatures and data formats, and the adapter bridges them without modifying the legacy code.

3. Class Diagram



Q4



Q5 (a)

Boundary (User / Interaction Points)

WebDashboard – Interface for operators to deploy services and monitor status

Monitoring Inputs – Streams performance data to the system (CPU, health)

Control (Decision-Making / Coordination)

Scaling Logic – Decides when and how many instances to launch or remove

Recovery Logic – Decides when to restart failed instances

Deployment Management Logic – Coordinates changes to service configurations

Entity (Stored State / Passive Data)

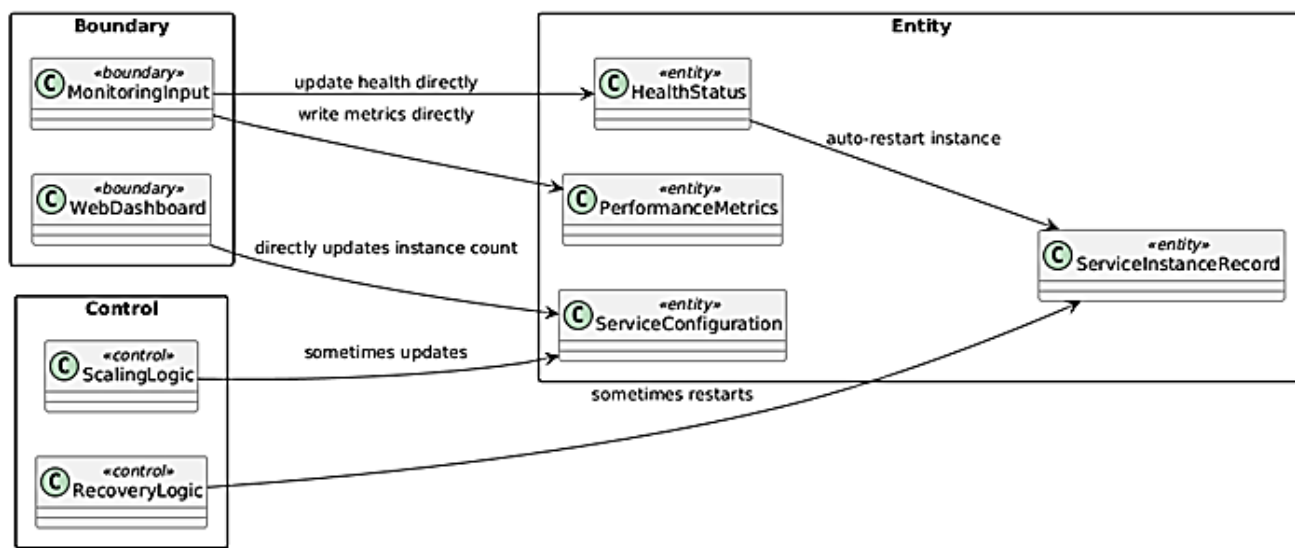
ServiceConfiguration – Desired number of instances, deployment settings

PerformanceMetrics – CPU usage, resource consumption

HealthStatus – Running or failed state for each instance

Service Instance Records – Tracks actual running instances

Q5 (b)

Incorrect ECB Design - Cloud Service Scaling System

Q5 (c) :

Incorrect Relationship 1 : WebDashboard → ServiceConfiguration: The boundary (WebDashboard) **directly updates the persistent state**, bypassing control logic.

Issue Observed: Instance count displayed on the dashboard does not match actual running instances; scaling happens inconsistently.

Incorrect Relationship 2 : HealthStatus → Automatic Restart: The entity (HealthStatus) **initiates actions**, instead of being passive.

Issue Observed: Services restart repeatedly without operator action or decision logic; debugging is difficult.

Q5 (d)

Correct ECB Design - Cloud Service Scaling System

